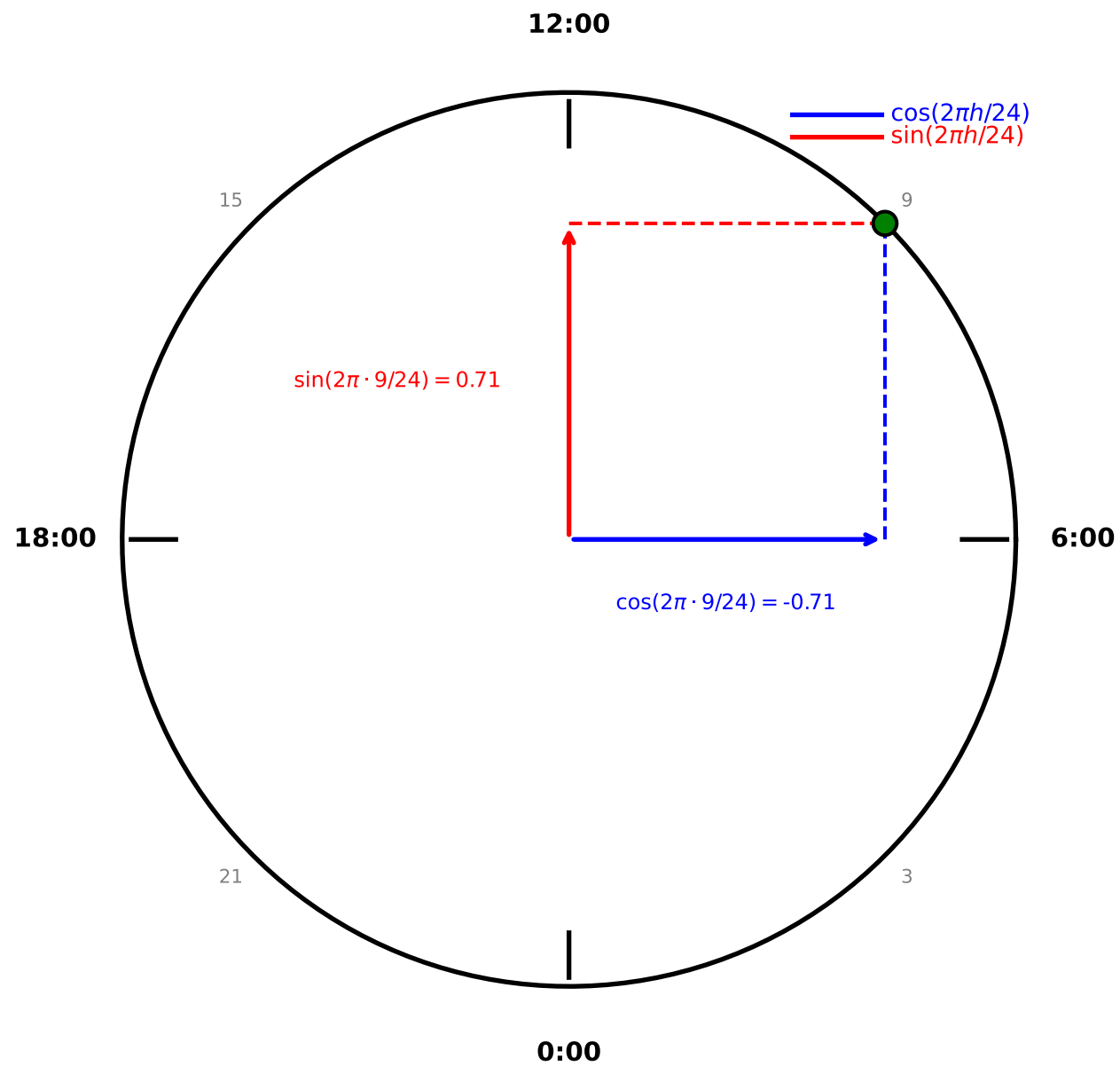


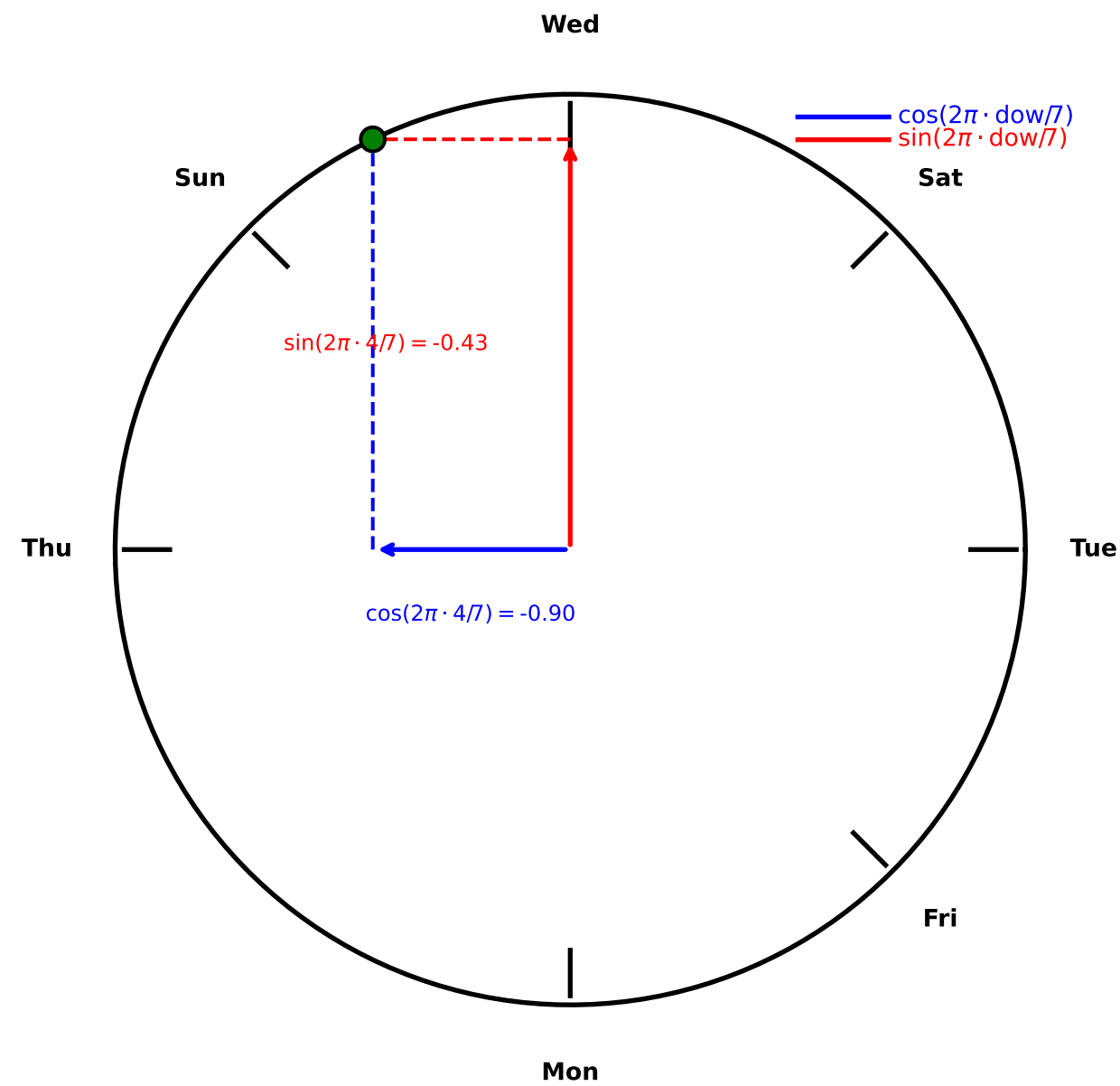
Cyclic Encoding for Temporal Features in CA-DQStream

Cyclic Encoding: Hour of Day (h=9)



hour_sin = $\sin(2\pi h/24)$
hour_cos = $\cos(2\pi h/24)$
Captures 23:59 $\approx$ 00:01

Cyclic Encoding: Day of Week (Friday)



dow_sin = $\sin(2\pi \cdot \text{dow}/7)$
dow_cos = $\cos(2\pi \cdot \text{dow}/7)$
Ensures Sunday $\approx$ Monday proximity